

STANDARD OPERATING PROCEDURE

DATABASE MIGRATION AND PORTING

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
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INTRODUCTION

Data migration, porting, and integration are critical processes in the life cycle of information systems, enabling organizations to move data from legacy systems to modern platforms, integrate diverse data sources, and ensure seamless operation of business applications. This SOP provides a structured approach to these activities, ensuring data integrity, security, and compliance throughout the process.

PURPOSE

The purpose of this SOP is to outline the standard procedures and best practices for planning, executing, and validating data migration, porting, and integration projects. The goal is to ensure these processes are carried out efficiently, securely, and in compliance with relevant regulations, thereby minimizing risks and disruptions to business operations.

SCOPE

This SOP applies to all data migration, porting, and integration projects within the organization, regardless of the size or complexity.

A. Data Porting :-

Applicable when any external data or data from legacy system need to be imported in the software system.

1. Understand and Identify :-

- 1.1 Understand relationships between data entities, such as foreign key constraints, joins, and lookups. Check and verify if all data related to application is considered. Ensure all required data is migrated to maintain application functionality.

- 1.2 Get the details of source and destination databases. Identify gaps and plan for necessary adjustments to ensure a smooth migration/porting.
- 1.3 Get the comprehensive understanding of the existing data structure and format (Flat file/ Spreadsheet/ DBMS etc) in the source database.
- 1.4 Document the structure of tables and columns, including data types, constraints (e.g., primary keys, foreign keys), and indexes.
- 1.5 Calculate the total size of the data to be ported.
- 1.6 Define the migration/porting strategy of porting all data at once or in incremental manner. Migrate data in phases to reduce risk and manage large volumes. This can include moving data by table, by data partition.

2. Source and Target Analysis :-

After understanding the structure and characteristics, analyze details of the source and target systems.

2.1 Source System Analysis:-

- i. Document the existing data schema (tables, columns, data types).
- ii. Analyze data quality and identify potential issues (e.g., duplicates, missing data, defaults, merged data).
- iii. Document inconsistency in data in the source data and bring it to the notice of PM to take up with the client and decide mitigation.

2.2 Target System Analysis :-

- i. Define the target data schema and structure.
- ii. Understand the target data/table structure/constraints.

3. Data Mapping and Transformation Design :-

3.1 Data Mapping :-

- i. Create a detailed mapping document that matches source data elements to target data elements.
- ii. Define data format changes, data cleansing, and data normalization.
- iii. Develop scripts or use ETL (Extract, Transform, Load) tools to automate transformations.

3.2 Data Validation Rules :-

- i. Establish validation rules to ensure data accuracy and consistency.

- ii. Plan for data integrity checks post-migration/porting.
e.g. group on master columns to see if any inconsistency. Validate it with the source data.

B. Data Migration :-

Applicable when data needs to be removed from source location to a different location between production systems and also may be due to version upgrades or data center relocation.

Carry out first two steps as defined for Data Porting.

4. Migration Plan :-

- 4.1 Decide if migration will be done all at once or in incremental approach.
- 4.2 Choose migration time for the downtime in consultation with PM and add it to the schedule.
- 4.3 Prepare migration schedule.
- 4.4 Understand if any external teams are needed during migration , such as Hardware team, implementation engineer etc.
- 4.5 If external team is needed plan for the respective responsibilities in the migration plan.

5. Pilot Migration :-

- 5.1 Conduct a pilot migration with data (limited data) to validate the design and identify potential issues.
- 5.2 Perform the pilot migration in a controlled environment.
- 5.3 Monitor performance and log any issues encountered. Log difference in time taken and validate with plan.
- 5.4 Validate the migrated data against the defined criteria.
- 5.5 Whenever deviation noticed then revise the plan.

6. Full Migration Execution :-

- 6.1 Before the planned migration date, brief the whole team on the responsibilities once again and share the migration schedule tracker sheet
- 6.2 Ensure all stakeholders are informed and prepared.
- 6.3 Note table row count of source location.
- 6.4 On the migration day start the activity as planned.

- 6.5 Stop the application as per the schedule to avoid data growth during downtime.
- 6.6 Backup all source data and ensure data recovery plans are in place.
- 6.7 Perform data extraction, transformation, and loading according to the plan or direct load transferred data.
- 6.8 Monitor the process in real-time and address any issues immediately.
- 6.9 During the course of migration keep updating the status on migration in the tracker scheduled.
- 6.10 Conduct thorough data validation and integrity checks. Check table row count before and after migration of the data.
- 6.11 Verify that all data has been migrated correctly and completely by validating the Data Validation Rules.
- 6.12 Test the performance of the target system with the full data set.
- 6.13 Optimize as needed based on test results.

7. Review and Optimization :-

- 7.1 Review the migration process and optimize for future projects.
- 7.2 Conduct a detailed review of the entire migration process.
- 7.3 Identify successes, challenges, and lessons learned.
- 7.4 Document insights and recommendations for future migrations.
- 7.5 Update the migration strategy and procedures based on the review.
- 7.6 Plan for future improvements and optimizations.
- 7.7 Conduct quarterly evaluation of the operation management and data porting with System Architect And CTO.